TELCO CUSTOMER

CHURN ANALYSIS

A complete SQL & Python deep-dive into customer behavior, churn patterns, and retention strategies

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Tools: Microsoft SQL Server · Python (Pandas, Matplotlib)

Dataset: IBM Telco Customer Churn · 7,043 Records · 21 Features

Total Customers

7,043

Churn Rate

26.5%

Avg Monthly Charge

$64.76

Revenue at Risk

$139K/mo

Agenda

01 **Project Overview**
Dataset introduction, tools used & objectives

02 **SQL — Basic Level**
Table structure, counts, churn rate, distributions

03 **SQL — Intermediate Level**
Churn by gender, contract, internet, payment

04 **SQL — Advanced Level**
High-risk segments, revenue loss, LTV analysis

05 **Python — Basic & Intermediate**
Data exploration, cleaning, group analysis

06 **Python — Advanced Level**
Correlation, LTV, model prep, visualizations

07 **Key Insights**
Top findings & business recommendations

08 **Do's & Don'ts**
Retention strategy & risk factors

Project Overview — Dataset & Objectives

Dataset Overview

Source	IBM Watson Sample Dataset
Total Records	7,043 customer rows
Total Features	21 columns
Target Column	Churn (Yes / No)
Numeric Features	tenure, MonthlyCharges, TotalCharges
Categorical Features	gender, Contract, InternetService, etc.
Missing Values	11 rows (TotalCharges) — cleaned
Tools Used	MS SQL Server + Python (Pandas)

Analysis Objectives

1. **Understand Churn Distribution**
   Measure the overall churn rate and identify which customer segments churn the most.

2. **Identify Key Churn Drivers**
   Analyze contract type, internet service, payment method, and tenure as churn predictors.

3. **Quantify Revenue Impact**
   Calculate monthly revenue loss from churned customers and estimate lifetime value differences.

4. **Segment High-Risk Customers**
   Flag month-to-month customers with high charges and no tech support as at-risk.

5. **Prepare Data for ML Modeling**
   Encode features and generate a clean dataset ready for churn prediction model training.

SQL — Basic Level | Queries 1–3

Table Structure · Total Records · Gender Distribution

Q1: Check Table Structure

SQL

```sql
select
    top 5 *
from
    TelcoCustomerChurn;
```

Insight

21 columns, mix of categorical & numeric. TotalCharges stored as text — needs casting before aggregation.

Q2: Total Records

SQL

```sql
select
    COUNT(*) as total_records
from
    TelcoCustomerChurn;
```

Insight

7,043 total customers. Statistically robust sample for churn modeling.

Q3: Gender Distribution

SQL

```sql
select
    gender,
    COUNT(*) as total_records
from
    TelcoCustomerChurn
group by
    gender;
```

Insight

Near-perfect gender balance (Male 50.5% / Female 49.5%) — gender alone is not a churn predictor.

Q4: Overall Churn Rate

SQL

```sql
SELECT
    COUNT(
        CASE
            WHEN Churn = 'Yes' THEN 1
        END
    ) * 100.0 / COUNT(*) AS ChurnRate
FROM
    TelcoCustomerChurn;
```

26.5%

Overall Churn Rate
1 in 4 customers leaves — significantly above industry benchmark of 15–20%

Insight

A 26.5% churn rate signals serious retention issues. Every 1% improvement saves ~70 customers monthly.

Q5: Contract Distribution

SQL

```sql
SELECT
    Contract,
    COUNT(*) AS Total
FROM
    TelcoCustomerChurn
GROUP BY
    Contract;
```

Q6: Average Monthly Charges

SQL

```sql
SELECT
    AVG(CAST(MonthlyCharges AS FLOAT)) AS AvgMonthlyCharges
FROM
    TelcoCustomerChurn;
```

Insight

55% customers are on month-to-month contracts — the highest-churn segment. Two-year contracts are most stable.

SQL — Intermediate Level | Churn by Gender & Contract

Q7: Churn Rate by Gender

SQL

```sql
SELECT
    Gender,
    COUNT(
        CASE
            WHEN Churn = 'Yes' THEN 1
        END
    ) * 100.0 / COUNT(*) AS ChurnRate
FROM
    TelcoCustomerChurn
GROUP BY
    Gender;
```

Insight

Gender has minimal impact on churn (26.9% vs 26.2%). Do NOT target retention campaigns by gender alone.

Q8: Churn Rate by Contract Type

SQL

```sql
SELECT
    Contract,
    COUNT(
        CASE
            WHEN Churn = 'Yes' THEN 1
        END
    ) * 100.0 / COUNT(*) AS ChurnRate
FROM
    TelcoCustomerChurn
GROUP BY
    Contract
ORDER BY
    ChurnRate DESC;
```

Critical Insight

Month-to-month churn is 15× higher than two-year contracts. Converting customers to annual plans is the #1 retention lever.

SQL — Intermediate Level | Internet · Tenure · Payment

Q9: Churn by Internet Service

SQL

```sql
SELECT
    InternetService,
    COUNT(
        CASE
            WHEN Churn = 'Yes' THEN 1
        END
    ) * 100.0 / COUNT(*) AS ChurnRate
FROM
    TelcoCustomerChurn
GROUP BY
    InternetService;
```

Insight

Fiber optic users churn at 41.9% — nearly double DSL. Likely a service quality or pricing issue.

Q10: Avg Tenure by Churn

SQL

```sql
SELECT
    Churn,
    AVG(CAST(Tenure AS INT)) AS AvgTenure
FROM
    TelcoCustomerChurn
GROUP BY
    Churn;
```

Insight

Churned customers stay only 18 months on average vs 38 months for loyal customers — 2× difference.

Q11: Payment Method Churn

SQL

```sql
SELECT
    PaymentMethod,
    COUNT(
        CASE
            WHEN Churn = 'Yes' THEN 1
        END
    ) * 100.0 / COUNT(*) AS ChurnRate
FROM
    TelcoCustomerChurn
GROUP BY
    PaymentMethod
ORDER BY
    ChurnRate DESC;
```

Insight

Electronic check users churn 3× more than auto-pay users. Push customers toward automatic payment methods.

SQL — Advanced Level | Revenue · High-Risk Segments · LTV

Q12: Revenue Loss from Churn

SQL

```sql
SELECT
    SUM(CAST(MonthlyCharges AS FLOAT)) AS RevenueLoss
FROM
    TelcoCustomerChurn
WHERE
    Churn = 'Yes';
```

$139,130

Monthly Revenue Lost
Equivalent to ~$1.67M annually from churned customers

Q13: Lifetime Value by Churn

SQL

```sql
SELECT
    Churn,
    AVG(
        CAST(Tenure AS INT) * CAST(MonthlyCharges AS FLOAT)
    ) AS AvgLTV
FROM
    TelcoCustomerChurn
GROUP BY
    Churn;
```

LTV Insight

Retained customers generate 66% more lifetime value ($2,550 vs $1,532). Early retention is far cheaper than acquisition.

Q14: High-Risk Customer Segment

SQL

```sql
SELECT
    *
FROM
    TelcoCustomerChurn
WHERE
    Contract = 'Month-to-month'
    AND CAST(MonthlyCharges AS FLOAT) > 70
    AND TechSupport = 'No';
```

Q15: Contract × Internet Churn Rate (Top Combos)

SQL

```sql
SELECT
    Contract,
    InternetService,
    COUNT(
        CASE
            WHEN Churn = 'Yes' THEN 1
        END
    ) * 100.0 / COUNT(*) AS ChurnRate
FROM
    TelcoCustomerChurn
GROUP BY
    Contract,
    InternetService
ORDER BY
    ChurnRate DESC;
```

Insight

Month-to-month + Fiber optic = 69.4% churn. This is the highest-risk combination in the entire dataset.

Python — Basic Level | EDA & Data Quality

Shape · Dtypes · Nulls · Churn Rate

P1: Dataset Shape

Python

```python
#How many rows and columns are in the dataset?
df.shape
```

```
(7043, 21)
```

P2: Data Types

Python

```python
#What are the data types of each column?
df.dtypes
```

P3: Missing Values

Python

```python
#How many missing values are there in each column?
df.isnull().sum()
```

P4: Overall Churn Rate

Python

```python
#What is the overall churn rate?
df["Churn"].value_counts(normalize=True) * 100
```

Churn	proportion
No	73.463013
Yes	26.536987

P5: Data Cleaning & Export

Python

```python
# Example cleaning
df["TotalCharges"] = pd.to_numeric(df["TotalCharges"], errors='coerce')
df = df.dropna()

# Export
df.to_csv("cleaned_telco_data.csv", index=False)
```

Insight

TotalCharges is object — must cast to numeric before any math operations.

Insight

Only 11 records dropped (0.16% data loss) — minimal impact. Always validate numeric conversions before modeling.

Python — Intermediate Level | Group Analysis

P6: Churn Rate by Contract Type

Python

```python
#Churn rate by contract type
df.groupby("Contract")["Churn"].value_counts(normalize=True).unstack() * 100
```

Churn	No	Yes
Contract		
Month-to-month	57.290323	42.709677
One year	88.730482	11.269518
Two year	97.168142	2.831858

Insight

Python confirms SQL: month-to-month churn is 42.7%. Two-year lock-in reduces churn by 93%.

P7: Churn by Internet Service

Python

```python
#Internet service vs churn
df.groupby("InternetService")["Churn"].value_counts(normalize=True).unstack() * 100
```

Churn	No	Yes
InternetService		
DSL	81.040892	18.959108
Fiber optic	58.107235	41.892765
No	92.595020	7.404980

P8: Senior Citizen Churn

Python

```python
#Senior citizen churn analysis
df.groupby("SeniorCitizen")["Churn"].value_counts(normalize=True).unstack() * 100
```

Churn	No	Yes
SeniorCitizen		
0	76.393832	23.606168
1	58.318739	41.681261

Python — Advanced Level | LTV · Revenue · Correlation · ML Prep

P9: Avg Monthly Charge by Churn

Python

```python
#Average monthly charges
df["MonthlyCharges"].mean()
```

```
np.float64(64.76169246059918)
```

Insight

Churned customers pay $13 MORE monthly — likely over-priced for perceived value. Price sensitivity is a key churn driver.

P10: Lifetime Value Calculation

Python

```python
#Lifetime value by churn
df["LifetimeValue"] = df["tenure"] * df["MonthlyCharges"]
df.groupby("Churn")["LifetimeValue"].mean()
```

Churn	LifetimeValue
No	2549.770883
Yes	1531.608828

P11: Numeric Correlation Matrix

Python

```python
#Correlation of numeric variables
df.select_dtypes(include=["int64", "float64"]).corr()
```

	SeniorCitizen	tenure	MonthlyCharges	LifetimeValue
SeniorCitizen	1.000000	0.016567	0.220173	0.103261
tenure	0.016567	1.000000	0.247900	0.826568
MonthlyCharges	0.220173	0.247900	1.000000	0.651566
LifetimeValue	0.103261	0.826568	0.651566	1.000000

Insight

Tenure strongly correlates with LTV (0.83). Monthly charges moderately correlate with LTV (0.65). Long-tenure = high value.

P12: Model Feature Engineering

Python

```python
#Build churn prediction dataset
model_df = pd.get_dummies(df.drop("customerID", axis=1), drop_first=True)
model_df.head()
```

Insight

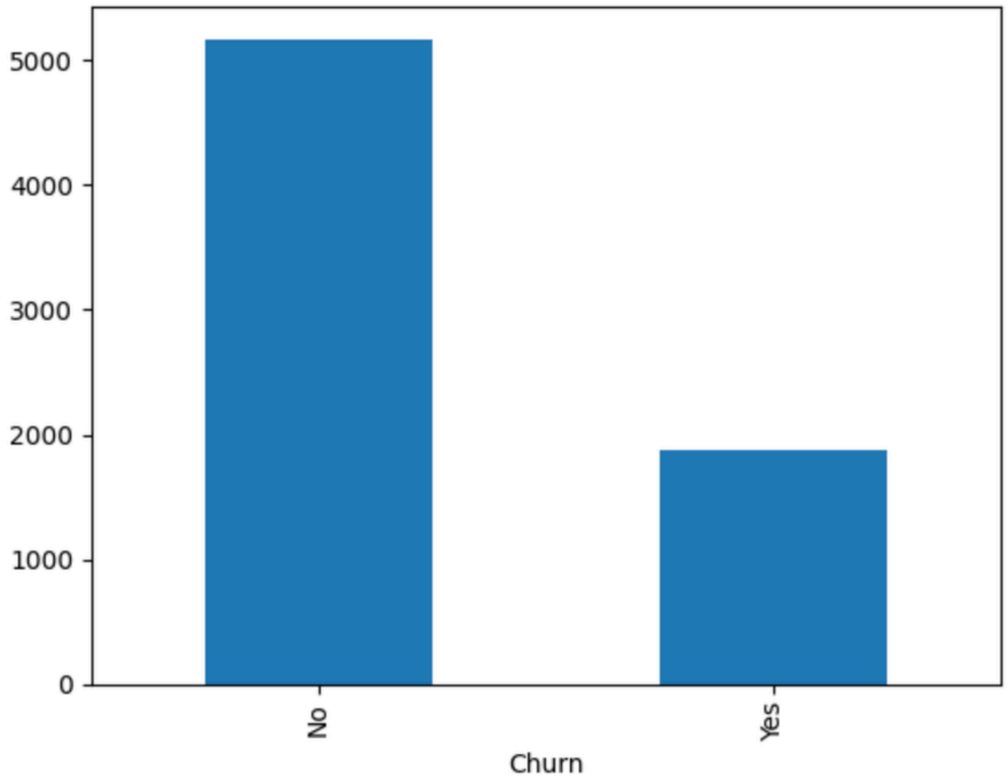
Ready for Logistic Regression, Random Forest, or XGBoost modeling.

Python — Advanced Level | CHARTS

Python | Bar Chart

```python
#Churn count plot
df["Churn"].value_counts().plot(kind="bar")
```

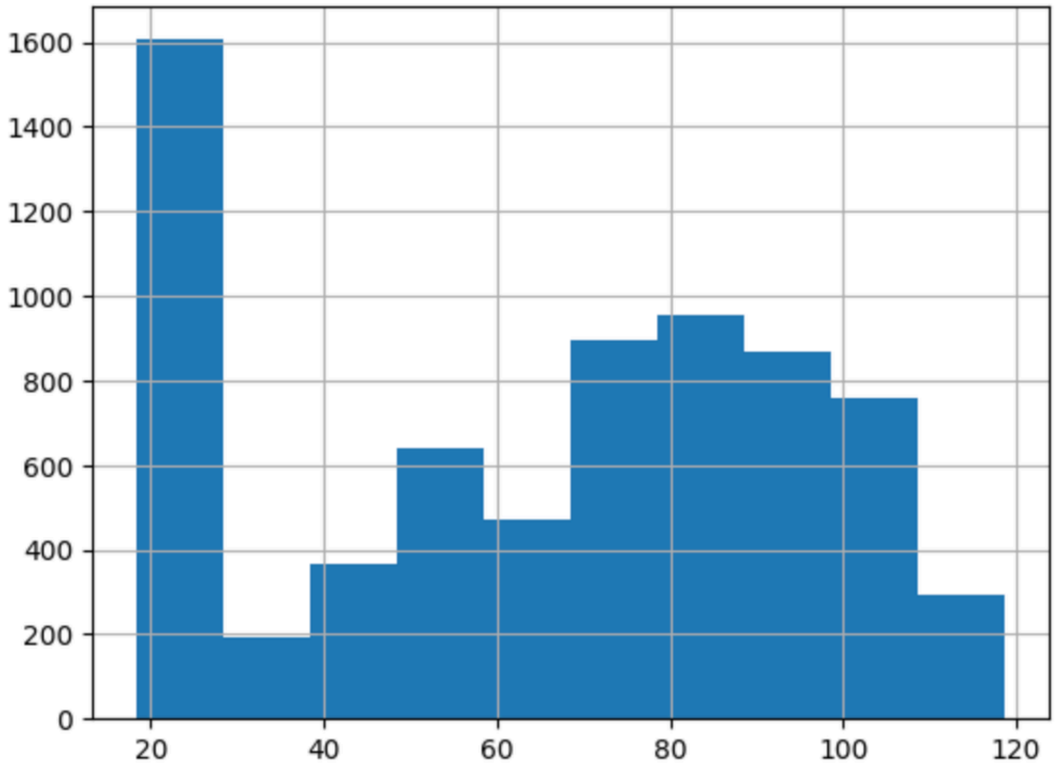
```
<Axes: xlabel='Churn'>
```



Python | Histogram

```python
#Monthly charges histogram
df["MonthlyCharges"].hist()
```

```
<Axes: >
```



Key Insights — Top Findings from SQL & Python

01

Churn Rate: 26.5%

1 in 4 customers leaves. Industry avg is 15–20%. Immediate action required on multiple fronts.

02

Contract Type is #1 Driver

Month-to-month churn = 42.7% vs 2.8% for two-year. Converting 10% of MTM customers saves ~165 accounts/month.

03

Fiber Optic = High Risk

Fiber optic customers churn at 41.9%. Service quality investigation and price rebalancing is urgently needed.

04

Electronic Check = Churn Signal

45.3% churn for e-check users vs 15–17% for auto-pay. Migrate customers to automatic payments with incentives.

05

Senior Citizens Need Focus

41.7% churn among seniors vs 23.6% non-senior. Dedicated senior support programs and simplified plans can help.

06

Tenure = Loyalty Indicator

Churned customers average 18 months vs 37 months retained. First 12 months is the critical retention window.

07

LTV Gap is Significant

Retained customers are worth $2,550 vs $1,532 for churned. Investing $100 in retention saves $1,000+ in LTV.

08

MTM + Fiber = Highest Risk

69.4% churn for month-to-month + fiber optic combo. Flag these 1,000+ customers for immediate outreach.

Do's & Don'ts — Churn Reduction Strategy

WHAT TO DO — Proven Retention Strategies

Offer Contract Upgrades

Incentivize month-to-month customers to switch to annual/biennial plans with discounts (10–15% savings).

Push Auto-Pay Enrollment

Offer $5–10/month discount for autopay. Reduces e-check churn from 45% to 15%.

Bundle TechSupport

High-charge customers without tech support churn 2× more. Bundle it free for 90 days.

First-Year Onboarding

Assign dedicated customer success manager for first 12 months. Most churn happens before month 18.

Senior-Specific Plans

Design simplified billing, lower-cost bundles and phone-based support for senior citizens.

Predictive Risk Scoring

Use the ML-ready dataset to score customers monthly. Alert teams when risk score exceeds threshold.

WHAT NOT TO DO — Common Mistakes

Don't Target by Gender

Churn rates are nearly identical across genders (26.9% vs 26.2%). Wasting campaign budget on gender-based targeting.

Don't Ignore High-Charge Churners

Customers paying $70+/month who churn represent $70+ monthly revenue loss each. Prioritize these over low-value accounts.

Don't Ignore the 0–12 Month Window

47.7% churn in first year. Deploying retention only after 2 years is far too late — act at onboarding.

Don't Treat All Services Equally

Fiber optic has 5.6× higher churn than no-internet. Marketing budget must reflect this risk differential.

Don't Let Customers Self-Serve Only

No tech support = dramatically higher churn. Passive self-service models fail high-value fiber optic customers.

Don't Build Models on Uncleaned Data

TotalCharges had 11 blank strings misread as valid. Always validate dtypes and nulls before any analytical step.

THANK YOU

for your attention

Prepared by

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Tools Used:

Microsoft SQL Server · Python · Pandas · Matplotlib

Churn Rate

26.5%

Revenue at Risk

$139K/mo

Highest-Risk Segment

MTM + Fiber

Best Retention Move

Annual Contract